MAKOTO M. KELP

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EDUCATION

Harvard University 2018 – 2023
Ph.D., Earth and Planetary Sciences
S.M., Environmental Science and Engineering

Reed College 2012 – 2016
B.A., Chemistry

PROFESSIONAL EXPERIENCE

Assistant Professor, University of Utah Starting Jan 2026
Department of Atmospheric Sciences, Wilkes Center for Climate Science & Policy

NOAA Climate & Global Change Postdoctoral Fellow, Stanford University 2023 – Present
Host: Noah Diffenbaugh

Graduate Research Assistant, Harvard University 2018 – 2023
Advisors: Daniel Jacob and Loretta Mickley

Junior Research Scientist, University of Washington 2016 – 2018
Advisor: Julian Marshall

RESEARCH INTERESTS

My research centers on applying machine learning and data-driven methods to uncover new perspectives in atmospheric chemistry and land-climate-human interactions. I place a special emphasis on exploring the interplay among fires, climate, and society.

AWARDS AND FELLOWSHIPS

NOAA Climate & Global Change Postdoctoral Fellowship 2023 – 2025
Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVII) 2023
Harvard University Bok Center Certificate of Distinction in Teaching 2019, 2022
AGU Outstanding Student Presentation Award 2019
National Science Foundation STEM Scholar 2013 – 2016

PUBLICATIONS (*SUBMITTED, <u>ADVISEE</u>)

h-index: 12 (Google Scholar)

*24. <u>Chung, K.</u>, T. Liu, **M. Kelp**, K. Vohra, D. Skelly, M. Carroll, J. Schwartz, and L.J. Mickley. Managing Smoke Risk from Wildland Fires: Northern California as a Case Study. (Submitted)
Co-advised undergraduate first-author with T. Liu

*23. Feng, X., L.J. Mickley, J.O. Kaplan, **M. Kelp**, Y. Li, T. Liu. Large role of anthropogenic climate change in driving smoke exposure across the western United States from 1992 to 2020. (Submitted)

*22. Hickman, S., **M. Kelp**, P. Griffiths, K. Doerksen, K. Miyazaki, E. Pennington, G. Koren, F. Iglesias-Suarez, M. Schultz, K. Chang, O.R. Cooper, A. Archibald, R. Sommariva, D. Carlson, H. Wang, J. West, and Z. Liu. Applications of Machine Learning and Artificial Intelligence in Tropospheric Ozone Research. (Submitted)
Tropospheric Ozone Assessment Report Phase II (TOAR-II) Community Special Issue

*21. **Kelp, M.**, M. Burke, M. Qiu, I. Higuera-Mendieta, T. Liu, and N. Diffenbaugh. Efficacy of Recent Prescribed Burning and Land Management on Wildfire Burn Severity and Smoke Emissions in the Western United States. (In review)

*20. Qiu, M., J. Li, C.F. Gould, R. Jing, **M. Kelp**, M.L. Childs, J. Wen, Y. Xie, M. Lin, M.V. Kiang, S. Heft-Neal, N.S. Diffenbaugh, M. Burke. Wildfire smoke exposure and mortality burden in the US under future climate change. (In review)

19. Qiu, M., D. Chen, **M. Kelp**, J. Li, G. Huang, M.D. Yazdi (2025). The rising threats of wildland-urban interface fires in the era of climate change: The Los Angeles 2025 fires. *The Innovation*, DOI: https://doi.org/10.1016/j.xinn.2025.100835.

18. Kawano, A., **M. Kelp**, M. Qiu, K. Singh, E. Chaturvedi, I. Azevedo, and M. Burke (2025). Improved daily $PM_{2.5}$ estimates in India reveal inequalities in recent enhancement of air quality. *Science Advances*, 11, 4, DOI: sciadv.adq1071.

17. Qiu, M., **M. Kelp**, S. Heft-Neal, X. Jin, C.F. Gould, D.Q. Tong, and M. Burke (2024). Evaluating Estimation Methods for Wildfire Smoke and their Implications for Assessing Health Effects. *Environ. Sci. Technol.*, 58, 52, 22880–22893, DOI: 10.1021/acs.est.4c05922.

    Special Issue on "Wildland Fires: Emissions, Chemistry, Contamination, Climate, and Human Health"

16. Liu, T., F.M. Panday, M.C. Caine, **M. Kelp**, D.C. Pendergrass, and L.J. Mickley (2024). Is the smoke aloft? Caveats regarding the use of the Hazard Mapping System (HMS) smoke product as a proxy for surface smoke presence across the United States. *International Journal of Wildland Fire*, 33, WF23148, DOI: 10.1071/WF23148.

15. Lin, H., L.K. Emmons, E.W. Lundgren, L.H. Yang, X. Feng, R. Dang, S. Zhai, Y. Tang, **M. Kelp**, N.K. Colombi, S.D. Eastham, T.M. Fritz, A.M. Fiore, and D.J. Jacob (2024). Intercomparison of GEOS-Chem and CAM-chem tropospheric oxidant chemistry within the Community Earth System Model version 2 (CESM2). *Atmospheric Chemistry and Physics*, 24, 8607–8624, DOI: 10.5194/acp-24-8607-2024.

14. **Kelp, M.**, C. A. Keller, K. Wargan, B.M. Karpowicz, and D. J. Jacob (2023c). Tropospheric ozone data assimilation in the NASA GEOS Composition Forecast modeling system (GEOS-CF v2.0) using satellite data for ozone vertical profiles (MLS), total ozone columns (OMI), and thermal infrared radiances (AIRS, IASI). *Environ. Res. Lett.*, 18, 094036, DOI: 10.1088/1748-9326/acf0b7.

13. **Kelp, M.**, T. C. Fargiano, S. Lin, T. Liu, J.R. Turner, J. N. Kutz, and L.J. Mickley (2023b). Data-driven placement of $PM_{2.5}$ air quality sensors in the United States: an approach to target urban environmental injustice, *GeoHealth*, 7, e2023GH000834, DOI: 10.1029/2023GH000834.

    Special Collection on "Geospatial data applications for environmental justice"

12. Balasus, N., D. J. Jacob, A. Lorente, J. D. Maasakkers, R. J. Parker, H. Boesch, Z. Chen, **M. Kelp**, H. Nesser, and D. J. Varon (2023). A blended TROPOMI+GOSAT satellite data product for atmospheric methane using machine learning to correct retrieval biases. *Atmos. Meas. Tech.*, 16, 3787–3807, DOI: 10.5194/amt-16-3787-2023.

11. **Kelp, M.**, M. Carroll, T. Liu, R.M. Yantosca, H.E. Hockenberry, and L.J. Mickley (2023a). Prescribed burns as a tool to mitigate future wildfire smoke exposure: Lessons for states and environmental justice communities. *Earth's Future*, 11, e2022EF003468, DOI: 10.1029/2022EF003468.

10. **Kelp, M.**, D.J. Jacob, H. Lin, and M.P. Sulprizio (2022b). An online-learned neural network chemical solver for stable long-term global simulations of atmospheric chemistry. *JAMES*, 14, e2021MS002926, DOI: 10.1029/2021MS002926.
    Selected as Highlight Paper
    Special Collection on "Machine learning application to Earth system modeling"

9. Yang, L. H., D.H. Hagan, J.C. Rivera-Rios, **M. Kelp**, E.S. Cross, C.Y. Peng, J. Kaiser, L.R. Williams, P. L. Croteau, J.T. Jayne, and N.L. Ng (2022). Investigating the sources of urban air pollution using low-cost air quality sensors at an urban Atlanta site. *Environ. Sci. Technol.*, 56, 11, 7063–7073, DOI: 10.1021/acs.est.1c07005.

    Special Issue on "Urban Air Pollution and Human Health"

8. **Kelp, M.**, S. Lin, J.N. Kutz, and L.J. Mickley (2022a). A new approach for optimal placement of $PM_{2.5}$ air quality sensors: case study for the contiguous United States. *Environ. Res. Lett.*, 17, 034034, DOI: 10.1088/1748-9326/ac548f.

7. **Kelp, M.**, D.J. Jacob, J.N. Kutz, J.D. Marshall, and C.Tessum (2020b). Toward stable, general machine-learned models of the atmospheric chemical system. *JGR: Atmospheres*, 125, e2020JD032759, DOI: 10.1029/2020JD032759.

6. **Kelp, M.**, T. Gould, E. Austin, J.D. Marshall, M. Yost, C. Simpson, and T. Larson (2020a). Sensitivity analysis of area-wide, mobile source emission factors to high-emitter vehicles in Los Angeles. *Atmospheric Environment*, 223, 117212, DOI: 10.1016/j.atmosenv.2019.117212.

5. Wen, Y., H. Wang, T. Larson, **M. Kelp**, S. Zhang, Y. Wu, and J.D. Marshall (2019). On-highway vehicle emission factors, and spatial patterns, based on mobile monitoring and absolute principal component score. *Science of The Total Environment*, 676, 242-251, DOI: 10.1016/j.scitotenv.2019.04.185.

4. **Kelp, M.**, C. Tessum, and J.D. Marshall (2018b). Orders-of-magnitude speedup in atmospheric chemistry modeling through neural network-based emulation. *arXiv*:1808.03874.

3. **Kelp, M.**, A.P. Grieshop, C.O. Reynolds, J. Baumgartner, G. Jain, K. Sethuramanand, and J.D. Marshall (2018a). Real-time indoor measurement of health and climate-relevant air pollution concentrations during a carbon-finance-approved cookstove intervention in rural India. *Development Engineering*, 3, 125-132, DOI: 10.1016/j.deveng.2018.05.001.

2. Brewer, J. F., M. Bishop, **M. Kelp**, C. Keller, A.R. Ravishankara, and E.V. Fischer (2017). A sensitivity analysis of key factors in the modeled global acetone budget. *J. Geophys. Res.*, 122, DOI: 10.1002/2016JD025935.

1. Jaffe, D., J. Putz, G. Hof, G. Hof, J. Hee, D.A. Lommers-Johnson, F. Gabela, J. Fry, B. Ayres, **M. Kelp**, and M. Minsk (2015). Diesel particulate matter and coal dust from trains in the Columbia River Gorge, Washington state, USA. *Atmospheric Pollution Research*, 6, 946-952, DOI: 10.1016/j.apr.2015.04.004.

INVITED TALKS AND SEMINARS

2024 UC Irvine – Earth System Science
2024 NOAA Climate and Global Change Summer Institute
2024 MIT – Civil and Environmental Engineering
2024 University of Washington – Atmospheric Science and Computer Science
2024 University of Utah – Atmospheric Science
2024 AI for the Study of Environmental Risk (AI4ER) seminar, University of Cambridge
2023 Southwest Fire Science Consortium and Fire Adapted New Mexico Learning Network
2023 PSEG Institute for Sustainability Studies with The Trust for Public Land, Montclair State University
2023 Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS) XVII, Brookhaven National Lab
2023 NASA GISS
2023 Columbia University
2023 Science in the News, Harvard University
2023 AGU/AMS GeoHealth Showcase
2023 MIT Atmospheric Chemistry Colloquium
2023 Stanford University
2022 Royal Meteorological Society Atmospheric Chemistry Special Interest Conference
2022 Pennsylvania Department of Environmental Protection Air Monitoring Committee Workshop
2022 Karlsruhe Institute of Technology
2022 University of Washington
2022 University of Illinois at Urbana-Champaign Advanced Environmental Engineering Seminar
2022 EPA Model Applications Team Meeting

SELECTED CONFERENCE PRESENTATIONS (*INVITED)

2024 AGU Fall Meeting, Washington D.C. (Talk)
*2024 Health Effects Institute Annual Conference, Philadelphia, PA (Talk)
2023 AGU Fall Meeting, San Francisco, CA (Talk)
*2023 Meteorology and Climate - Modeling for Air Quality, Session: "Breakthrough Innovations in Atmospheric & Air Quality Modeling", UC Davis (Talk)

2023 AMS Annual Meeting, Denver, CO (Talk)
2022 AGU Fall Meeting, Chicago, IL (Talk)
*2022 Atmospheric Chemical Mechanisms, Session: "Mechanism Development and Reduction", UC Davis (Talk)
2022 AMS Annual Meeting, Virtual (Talk)
2022 10th International GEOS-Chem Meeting, Washington University in St. Louis (Talk)
*2022 ECMWF Machine Learning Workshop (Talk)
2021 AMS Annual Meeting, Virtual (Talk)
*2020 AGU Fall Meeting, Virtual (Talk)
2020 Atmospheric Chemical Mechanisms Conference, Virtual (Talk)
2019 AGU Fall Meeting, San Francisco, CA (Poster)
2018 AGU Fall Meeting, Washington D.C. (Poster)

TEACHING EXPERIENCE

Stanford University Civil and Environmental Engineering
CEE265F/SUSTAIN248: Environmental Governance and Climate Resilience

Guest Lecturer **Winter 2025**

Harvard University Department of Earth and Planetary Sciences
EPS 200: Graduate-level Atmospheric Chemistry and Physics

Teaching Fellow **Fall 2019, Fall 2020, Fall 2022**

Guest Lecturer **Fall 2022**

Derek Bok Center for Education & Learning

Teaching Certificate **2023**

Reed College Department of Chemistry
Chem 101: Molecular Structure and Properties
Chem 102: Chemical Reactivity
Chem 230: Environmental Chemistry

Laboratory Teaching Assistant **2015-2016**

Tutor, Grader **2013-2016**

RESEARCH MENTORING

Undergraduates:

- Jason Hu (Winter 2025 – present, Stanford University): *"Machine learning transformers for embedded atmospheric chemistry solvers"*
- Kenith Taukolo (Winter 2025 – present, Harvard University): *"Effectiveness of prescribed burning in the western US"*
- Rebecca Sandoval (Summer 2024, University of California, Santa Barbara): *"Evaluating prescribed fire emissions in California"*
- Lyna Kim (Spring 2024 –present, Stanford University): *"Physically interpretable machine learning for climate and weather forecasting"*
- Christian Chiu (Summer 2023 – Fall 2023, Harvard University): *"Uncovering latent VOC emissions and spatiotemporal drivers of urban ozone in changing NO_x regimes"*
- Karina Chung (Summer 2023 – Fall 2024, Harvard University): *"Google Earth Engine applications for wildfire smoke risk in the Western United States"*
- Greta Schultz (Summer 2023, University of Wisconsin-Madison): *"Emergency mobile monitoring for California wildfire smoke"*
- Timothy Fargiano (Summer 2022 – Fall 2022, Harvard University): *"Optimal placement of $PM_{2.5}$ air quality sensors in the US: An approach to target environmental injustice"*

- Margaret Schultz (January 2022 – December 2022, Harvard University): *"Real-time high-resolution downscaling of fine particulate matter ($PM_{2.5}$) air quality in the United States using machine learning"*, year-long environmental engineering senior thesis (senior project spotlight)
- Sanjna Kedia (Summer 2022, Harvard University): *"Machine learning for automated detection of wildfire smoke in the US"*
- Samuel Lin (Summer 2021- Fall 2021, Harvard University): *"Optimal air quality sensor placement in the United States"*
- Marie Panday (Summer 2021, University of Maryland): *"Trends in and Reconstruction of Smoke Days across the United States"*
- Kent Toshima (Summer 2020 - Summer 2021, Harvard University): *"Application of deep learning to detection of wildfire smoke in HMS over North America"*
- Miah Caine (Summer 2020 - Spring 2021, Harvard University): *"Agreement between the HMS Product and Ground-Level Smoke in the Pacific Northwest"*

Committees as member:

- Vanessa Sun (UU ATMOS), PhD Committee 2024-

SYNERGISTIC ACTIVITIES

International:

Co-Chair for Tropospheric Ozone Assessment Report, Phase II (TOAR-II) Machine Learning for Tropospheric Ozone (ML4O3) Working Group *(2023 – Present)*

Co-leader of Statistical Learning in Atmospheric Chemistry (SLAC) group, seminar series (2022 – Present)

Proposal reviewer for *NASA (2022, 2024), NSF (2023, 2024), Spark Climate Solutions (2024)*

Peer reviewer for *Atmospheric Chemistry and Physics, Environmental Research Letters, JAMES, GeoHealth, Environmental Science & Technology, Earth's Future, ACS Earth and Space Chemistry, ES&T Air, Geoscientific Model Development, Communications Earth & Environment, Environmental Research Communications, Atmospheric Pollution Research, Environmental Data Science, Environmental Science and Pollution Research*

National:

Co-Chair for Atmospheric Chemistry Gordon Research Seminar (GRS) *(Summer 2025)*

Technical Program Committee for Atmospheric Chemical Mechanisms conference *(Dec 4-6, 2024)*

Session Convener at AGU Fall Meeting:

- Data-Driven Methods for Quantifying Atmospheric Composition: Advances in Computation and Statistical Learning, co-convener *(2023, 2024)*
- Managing Air Quality and Public Health Under Climate Change: Health Impacts, Mitigation Policy, and Adaptation Strategies, co-convener *(2024)*
- Prescribed Fires and Land Management in North America, primary convener *(2023)*

OSPA Judge at AGU Fall Meeting (2023, 2024)

University, Departmental:

DEI Liaison on behalf of postdocs in the Stanford Earth System Science Department *(2023-2025)*

- Nominated for a 2024 Stanford Postdoc JEDI (Justice, Equity, Diversity and Inclusion) Champion award

Co-leader of Machine Learning & Data Science Subgroup, Harvard Atmospheric Chemistry Modeling Group (2021 – 2023)

Organizer Harvard EPS Department G2 Quals Committee (2020-2021)

Organizer Harvard EPS Department Visiting Scholar Lecture Series Committee (2018 – 2019)

Press:

- Online machine learning in GEOS-Chem: Editor's Highlight in JAMES
- Prescribed fires: Press release, Harvard Gazette, KCRA Sacramento, CBS Newspath, Missoulian